

Encoding Methods

Encoding is the technique of converting categorical variables into a numerical representation. ML algorithms are based on mathematical equations and they can only operate on numerical data.

Q2: What are the main types of categorical data?

A: There are three primary types:

- **Nominal Data:** Categories without inherent order (e.g., colors, cities, animal types) 4 5
- **Ordinal Data:** Categories with natural order or ranking (e.g., education levels, size ratings) 4 5
- **Binary Data:** Simple either/or categories (e.g., True/False, Yes/No) 5

Categorical Data Encoding

• **One-Hot Encoding :** One-hot encoding creates separate binary columns for each category in a feature. For a given row, the column corresponding to its category gets as '1' and all other new column get '0'.

use: nominal variables

Dummy Encoding

Dummy coding scheme is similar to one-hot encoding. This categorical data encoding method transforms the categorical variable into a set of binary variables (also known as dummy variables). In the case of one-hot encoding, for N categories in a variable, it uses N binary variables. The dummy encoding is a small improvement over one-hot-encoding. Dummy encoding uses N-1 features to represent N labels/categories.

One Hot Encoding:-

Ex:

D1	people watch campusx
D2	campusx watch capusx
D3	people write comment
D4	campusx write comment

Corpus

people watch campusx campusx watch capusx
people write comment campusx write comment

Vocabulary

people watch campusx write comment

D1

people	watch	campusx	write	comment
1	0	0	0	0
0	1	0	0	0
0	0	1	0	0

$$D1 = \begin{bmatrix} [1, 0, 0, 0, 0], \\ [0, 1, 0, 0, 0], \\ [0, 0, 1, 0, 0] \end{bmatrix}$$

Drawbacks:-

Curse of Dimensionality :-

- A large number of levels are present in data. If there are multiple categories in a feature variable in such a case we need a similar number of dummy variables to encode the data. For example, a column with 30 different values will require 30 new variables for coding.
- If we have multiple categorical features in the dataset similar situation will occur and again we will end to have several binary features each representing the categorical feature and their multiple categories e.g a dataset having 10 or more categorical columns.

In both the above cases, these two encoding schemes **introduce sparsity in the dataset** i.e several columns having 0s and a few of them having 1s. In other words, it creates multiple dummy features in the dataset without adding much information.

→ leads to overfitting.

Dummy variable Trap

One-Hot Encoding results in a Dummy Variable Trap as the outcome of one variable can easily be predicted with the help of the remaining variables. Dummy Variable Trap is a scenario in which variables are highly correlated to each other.

The Dummy Variable Trap leads to the problem known as [multicollinearity](#).

Label Encoding :- It assigns a unique integer to each category.

Ex: Red = 0, Blue = 1, Green = 2.

- Use: Ordinal data on tree-based models.

Drawbacks :-

Label encoding imposes an arbitrary order on categorical data, which can be misleading. In the given example, the countries have no inherent order, but one hot encoding and label encoding introduces an ordinal relationship based on the encoded integers (e.g., France < Germany < Spain). This can cause the model to falsely interpret these categories as having a meaningful order, potentially leading to incorrect inferences.

Target Encoding :- Each category is replaced with mean of the target variable for that category.

Ex: In loan prediction, each city might be replaced with that city's historical default rate.

- **Pros:** Can capture information from the target variable, often leading to better model performance. It doesn't increase the feature space.
- **Cons:** Highly prone to **overfitting**, especially for rare categories. Proper regularization or cross-validation strategies are needed to mitigate this. It can also cause **target leakage** if not implemented carefully.

Drawbacks of One Hot Encoding in NLP :

Cons:-

- Sparsity:- Each vector is almost entirely zeros except for a single 1. This wastes memory and computational resources. Sparsity leads to overfitting.
- No semantic information:-
Treats every word as orthogonal to every other.
"dog" and "puppy" are as unrelated as "dog" and "quantum".
- Out of Vocabulary Problem:-
Can't represent words not seen during vocabulary creation. These words are simply missing from encoding scheme. Requires special handling.

Encode Numerical Columns

Binning / Discretization

Discretization: It is the process of transforming continuous variables into categorical variables by creating a set of intervals, which are contiguous, that span over the range of the variable's values. It is also known as "**Binning**", where the bin is an analogous name for an interval.

Benefits :-

- (i) Handling Outliers:- Binning can reduce the impact of outliers. An usually large or small value gets placed into a bin with other values, so it's extreme nature it's contained.
- (ii) It helps capture non-linear relationships between features and the target variable.
- (iii) Reduces unique values for tree based models.

Disadvantages:

1. **Information loss:** Loses precision of original values
2. **Boundary issues:** Values near bin edges might be misclassified
3. **Subjectivity:** Choice of bin boundaries can be arbitrary
4. **Increased dimensionality:** If using one-hot encoding after binning
5. Risk of Overfitting: Supervised binning methods, if not used carefully, can leads to overfitting by creating bins that are too specific to the training data.

Types of Binning

(i) Unsupervised Binning:-

Equal width Binning (on Uniform Binning)

The algorithm divides the data into N intervals of equal size.

$$\text{Binwidth} = \frac{\text{max} - \text{min}}{N}$$

- simple and easy to implement.
- Can be highly skewed by outliers, leading to some bins having many data points while others have very few or none.

Equal Frequency Binning or (Quantile Binning)

The data is divided into bins such that each bin contains roughly the same no. of data points.

- **How it works:** It uses quantiles (like percentiles or deciles) to determine the bin edges. For example, to create 4 bins, you would use the 25th, 50th, and 75th percentiles as the dividers.
- **Pros:** Handles skewed data much better than equal-width binning.
- **Cons:** The bin widths can be very different, which might not be intuitive.

(c) **K-means binning:** This technique uses the clustering algorithm namely "K-Means Algorithm".

- This technique is mostly used when our data is in the form of clusters.

Custom binning: It is also known as “Domain” based binning. In this technique, you have domain knowledge about your business problem statement and by using your knowledge you have to do your custom binning.

For Example, We have an attribute of age with the following values

Age: 10, 11, 13, 14, 17, 19, 30, 31, 32, 38, 40, 42, 70, 72, 73, 75

Now after Binning, our data becomes:

Attribute	Age -1	Age -2	Age -3
After Binning	10, 11, 13, 14, 17, 19 Young	30, 31, 32, 38, 40, 42 Mature	70, 72, 73, 75 Old

(ii) Supervised Binning :-

This method uses the target variable to create bins, aiming to create intervals that are more predictive of the outcome.

- **Decision Tree Binning:** A decision tree is built with the continuous feature as the input and the target variable as the output. The splits in the tree provide the optimal bin boundaries. This is a very effective method as it directly considers the relationship with the target.

Q: How does binning a continuous variable affect a linear regression model?

- **A:** Binning a continuous variable essentially converts it into a categorical one. To use it in a linear regression model, you would typically need to one-hot encode the bins. This allows the model to capture non-linear relationships. For instance, if the relationship between age and income is U-shaped (higher for young and old, lower in the middle), a simple linear model would fail. However, by binning age, the model can assign different coefficients to each age group, effectively modeling this non-linear pattern. The downside is the loss of granular information.



Evaluation Metrics

class output: SVM, KNN

probability output: Logistic Regression, Random Forest, Boosting

Confusion matrix :-

A Confusion Matrix is a table that summarizes a classification model's performance by showing the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). It is the foundation for calculating metrics like precision, recall, and accuracy ¹.

• Confusion Matrix

		ACTUAL	
		INFECTED	HEALTHY
PREDICTED	INFECTED	TRUE POSITIVE	FALSE POSITIVE → Type I error
	HEALTHY	FALSE NEGATIVE ↓ Type II error	TRUE NEGATIVE

Accuracy: ratio of correct predictions to total no. of predictions.

$$\frac{TP + TN}{TP + TN + FP + FN}$$

use: Best used when the classes are balanced.

Precision: of all predictions I made for positive class, how many were actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

use: when cost of false positive is high.
For example in spam detection, we don't want to incorrectly classify an important email as spam.

(Sensitivity / True Positive Rate)

Recall: of all actual positive cases, how many did my model correctly identify.

$$\text{Recall} = \frac{TP}{TP + FN}$$

use: when cost of False Negative is high. For example in medical diagnosis for a serious disease, we want to identify all patients who actually have the disease.

F1-Score :- Harmonic mean of Precision and Recall.

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

use: A great all-around metric for imbalanced datasets when you care equally about minimizing both FP and FN.

F beta score :-

Some we would like to give more weight to either precision or recall.

$$F_1 = (1 + \beta^2) \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$$

Problem with Accuracy:-

Accuracy can be highly misleading for imbalanced datasets (where one class vastly outnumbers the other). A model can achieve high accuracy by simply always predicting the majority class, while failing to identify the minority class of interest. To address this, always look at metrics like Precision, Recall, and the F1-Score, which provide a more nuanced view of model performance, especially on the positive class ¹⁰.

- Imagine we always report back "healthy"

		ACTUAL	
		INFECTED	HEALTHY
PREDICTED	INFECTED	0	0
	HEALTHY	5	95

$$(0+95)/100 = 95\% \text{ Accuracy}$$

- Accuracy:
 - How often is the model correct?

95% accuracy for a model that always returns "healthy"!

Note: $\text{Recall} = \frac{0}{5} = 0$ alerts the model is not catching cases.

$$\text{Precision} = \frac{0}{0} = \text{undefined} \quad \text{"}$$

Trade-off between Precision and Recall:

A: Precision and Recall have an inverse relationship. If you increase one, you typically decrease the other. For example, if you want to increase recall (catch more sick patients), you might lower the threshold for diagnosis. This means you'll correctly identify more sick patients (higher recall), but you'll also misclassify more healthy patients as sick (lower precision). This trade-off is managed by adjusting the classification threshold.

Predict positive if output probability > 0.3 *higher recall;*

Predict negative if output probability < 0.3 *lower precision*

Predict positive if output probability ≥ 0.9 *higher precision;*

lower recall

Predict negative if output probability < 0.9

specificity :-

$$\text{specificity} = \frac{TN}{TN + FP}$$

False Positive Rate :-

$$1 - \text{specificity} = \frac{FP}{TN + FP}$$

Receiver operating characteristic curve (ROC) :

The ROC curve is a graph that shows a model's performance across all classification thresholds.

It plots the True positive rate against the False positive Rate.

AUC (Area Under ROC Curve) :-

It provides aggregate measure of performance across all classification thresholds.

$$AUC = 1$$

- X-axis : True Positive Rate

- Y-axis : False Positive Rate

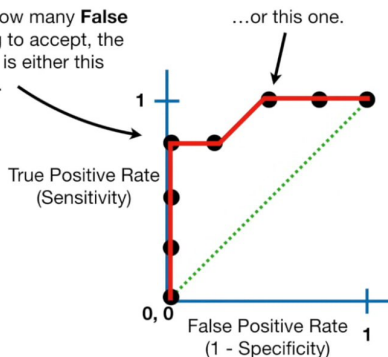
2. Important Points on ROC Curve:

- (0,0): Never predicts positive (conservative threshold)
- (1,1): Always predicts positive (liberal threshold)
- (0,1): Perfect classifier (ideal point)
- Diagonal line: Random guessing (no discriminative power)

• Interpretation:

- $AUC = 1$: Perfect classifier.
- $AUC = 0.5$: Useless classifier (equivalent to a random guess).
- $AUC < 0.5$: Worse than random.

And depending on how many **False Positives** I'm willing to accept, the optimal threshold is either this one...



- AUC = Probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance.

2. Threshold Selection:

- **Lower threshold**: More sensitive, higher TPR but also higher FPR
- **Higher threshold**: More specific, lower FPR but also lower TPR

Q4: What does an AUC score of 0.8 mean?

A: An AUC of 0.8 means there is an 80% chance that the model will rank a randomly chosen positive instance higher than a randomly chosen negative instance. In simpler terms, it indicates that the model has a good measure of separability and is effective at distinguishing between the positive and negative classes.

Q4: How do you choose the optimal threshold from ROC curve?

A: Methods include:

- **Youden's J statistic**: Maximize $(TPR - FPR)$
- **Cost-based approach**: Minimize total cost = $FN_cost \times FN + FP_cost \times FP$
- **Business constraints**: Fixed FPR requirement (e.g., max 5% false alarms)
- **Equal error rate**: Where $TPR = 1 - FPR$

Q. Can two models have the same AUC but different ROC curves?

This is a crucial concept because the *shape* of the ROC curve tells you which model is better for a specific business need.

Let's consider two models, A and B, both with an **AUC of 0.85**.

- **Model A** might be excellent at the very beginning, achieving a high TPR with an extremely low FPR. Its curve would be very steep on the left side. This is ideal for situations where you cannot tolerate false positives, like flagging legitimate bank transactions as fraudulent.
- **Model B** might have a more gradual, rounded curve. It might not be as good as Model A at very low FPRs, but it maintains a good balance across a wider range of thresholds. This could be useful in medical screening, where you want a generally good model that balances catching diseases with not alarming healthy patients.

$$\underline{TPR} = 1 \quad TP / (TP + FN)$$

$$FN = 1, TP = 1$$

$$+1 \Rightarrow 3$$

$$-1 \Rightarrow 7$$

$$\underline{r=4}$$

$$TP = 3$$

$$FN = 0$$

$$TP = 1$$

$$\underline{FPR}$$

$$FP = 4$$

$$TN = 3$$

$$FPR = 4/7$$

$$\underline{r=3}$$

$$FP =$$

$$P[z_i | x^i, p, q, c] = \sigma(z^i \cdot c^T x^i)$$

$$P[y_i | x^i, z_i=1, p, q, c] = \frac{1}{\sqrt{\pi}} e^{-(y_i - p^T x^i)^2}$$

$$P[y_i | x^i, p, q, c]$$

$$P[y_i | x^i, z_i=1, p, q, c] \cdot P[z_i=1 | x^i, \dots]$$

$$+ P[y_i | z_i=-1] P[z_i=-1]$$

$$\frac{1}{\sqrt{\pi}} e^{-(y^i - p^T x^i)^2} \cdot \sigma(c^T x^i) + \frac{1}{\sqrt{\pi}} e^{-(y^i - q^T x^i)^2} \cdot \sigma(-c^T x^i)$$

$$\sigma(c^T x^i) = \frac{1}{1 + e^{-c^T x^i}}, \quad \sigma(-c^T x^i) = \frac{1}{1 + e^{c^T x^i}}$$

$$\sum \ln [P(y_i | x_i, z_i, p, q, c)] + \underbrace{\sum \ln P(z_i | x_i, p, q, c)}$$

$$\sum \ln P[y_i | x_i, z_i, p, q, c]$$

$$\arg \min_{p, q} - \sum \ln P[y_i | x_i, z_i, p, q, c]$$

$$= \sum_{i=1}^n (y_i - q^T x_i)^2 + \sum_{i=1}^n (y_i - p^T x_i)^2$$

$$y_i = p^T x_i \quad y = Bx$$

$$(z_i = 1) \quad p^T = (x_{(1)}^T x_{(1)})^{-1} x_{(1)}^T y_{(1)}$$

$$z_i = -1 \quad q^T = (x_{(-1)}^T x_{(-1)})^{-1} x_{(-1)}^T y_{(-1)}$$

$$\sigma(z_i c^T x_i)$$

$$\ln \sum -\ln P[z_i | x_i, p, q, c]$$

$$= \sum + \ln (1 + e^{-z_i c^T x_i})$$







